

1. Consider the following set of processes (jobs), where the run time of the process is given in milliseconds (ms):

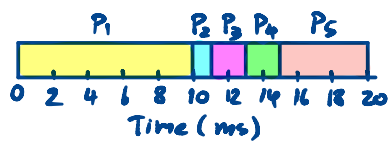
Process	Run Time
P_1	10
P_2	1
P_3	2
P_4	2
P_5	5

The processes are assumed to arrive in the order P_1, P_2, P_3, P_4, P_5 , all at time 0.

- a) Draw two diagrams illustrating the execution of these processes using FIFO, SJF, and RR scheduling with time slice = 1 ms.
b) What is the Turnaround time of each process for each of the schedulers in part (a)?

$$T_{\text{turnaround}} = T_{\text{completion}} - T_{\text{arrival}}$$

FIFO



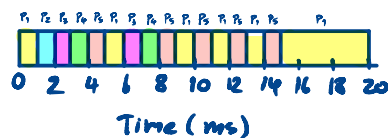
$$T = \frac{10 + 11 + 13 + 15 + 20}{5} = 13.8 \text{ ms} \#$$

SJF



$$T = \frac{20 + 10 + 5 + 3 + 1}{5} = 7.8 \text{ ms} \#$$

RR

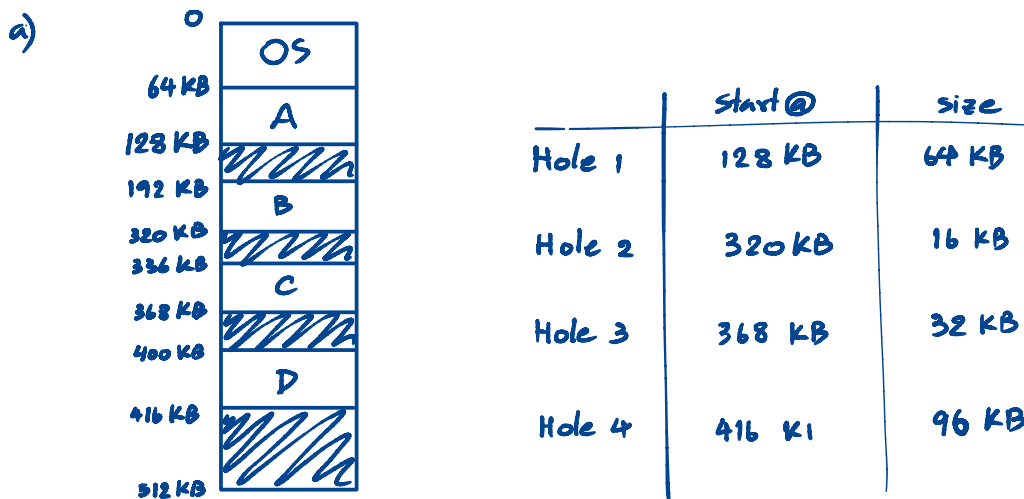


$$T = \frac{20 + 2 + 7 + 8 + 15}{5} = 10.4 \text{ ms} \#$$

2. A computer system has a total memory of 512 KB. It supports multiprogramming with contiguous memory allocation using flexible-size partitions. The operating system uses 64 KB memory, starting at address 0. At a given moment in time, the memory is occupied by four processes, A, B, C and D, as shown in table below

Process	Base Address	Size
A	64 KB	64 KB
B	192 KB	128 KB
C	336 KB	32 KB
D	400 KB	16 KB

- a) List the areas of free memory (holes) giving their start address and size.
b) Suppose the operating system receives a request from a new process E for 30 KB memory to be allocated. Assuming the free memory is searched in order of start address, what would be the base address of this new process using each of the following strategies: (i) First-Fit, (ii) Best-Fit, and (iii) Worst-Fit.



b) First-Fit : เลือก Hole แรกที่ว่าง
Base address is 128 KB (Hole 1)

Best-Fit : เลือก Hole ที่ว่างที่มีพื้นที่น้อยที่สุด
Base address is 368 KB (Hole 3)

Worst-Fit : เลือก Hole ที่ว่างที่มีพื้นที่มากที่สุด
Base address is 416 KB (Hole 4)

Question 2: (10 points)

6. (3 points) Shown below is the workload for 5 jobs arriving at time zero in the order given below

Job	Execution time
1	10
2	29
3	3
4	7
5	12

Calculate the average waiting time if three different algorithms are used: FCFS, SJF (non-preemptive), and Round Robin with time slice = 10.

- a) FCFS

ANS:

Job	waiting time
1	0
2	10
3	39
4	42
5	49



The average waiting time = $\frac{10 + 39 + 42 + 49}{5} = 28$

- b) SJF

ANS:

Job	waiting time
1	10
2	32
3	0
4	3
5	20



The average waiting time = $\frac{10 + 32 + 3 + 20}{5} = 13$

- c) RR

ANS:

Job	waiting time
1	0
2	61 - 29 = 32
3	23 - 3 = 20
4	40 - 7 = 33
5	52 - 12 = 40



The average waiting time = $\frac{32 + 20 + 33 + 40}{5} = 23$

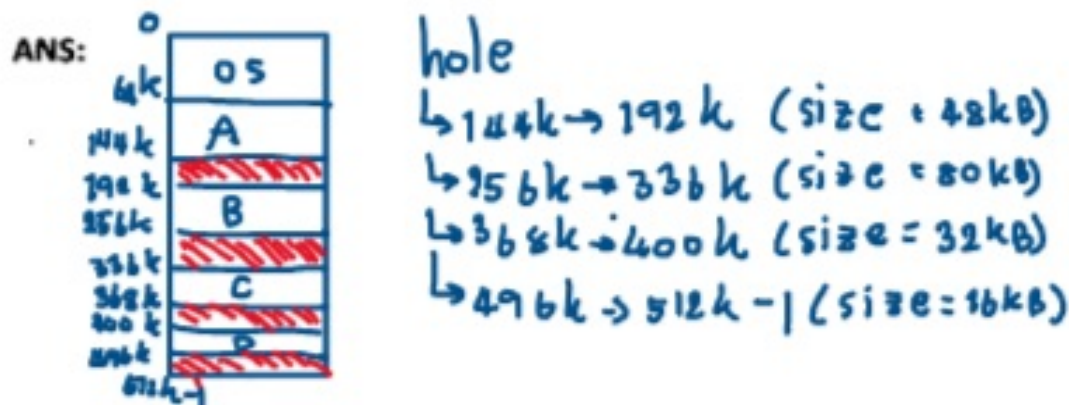
Question 4: (10 points)

A computer system has a total memory of 512 KB. It supports multiprogramming with contiguous memory allocation using flexible-size partitions. The operating system uses 64 KB memory, starting at address 0. At a given moment in time, the memory is occupied by four processes, A, B, C and D, as shown in Table Q4.

Process	Base Address	Size
A	64 KB	80 KB
B	192 KB	64 KB
C	336 KB	32 KB
D	400 KB	96 KB

Table Q4

- (a) (4 points) List the areas of free memory (holes) giving their start address and size.



- (b) (6 points) Suppose the operating system receives a request from a new process for 30 KB memory to be allocated. Assuming the free memory is searched in order of start address, what would be the base address of this new process using each of the following strategies (i) First-Fit, (ii) Best-Fit, and (iii) Worst-Fit. In each case, give a brief explanation of your answer (2 points each, 1 for answer, 1 for explanation).

ANS:

First-Fit: เลือก hole แรกที่บรรจุได้ $\rightarrow$ base = 144KB

Best-Fit: เลือก hole ที่ว่างเหลือพอขึ้นที่น้อยที่สุด $\rightarrow$ base = 368KB

Worst-Fit: เลือก hole ที่ว่างเหลือพอขึ้นที่มากที่สุด $\rightarrow$ base = 256KB